

**CHY1009**  
**Chemistry and Environmental Studies**

**Module 5**

**Nutrient Cycles**

## What Happens to Matter in an Ecosystem?

The elements and compounds that make up nutrients move continually through air, water, soil, rock, and living organisms within ecosystems, as well as in the biosphere in cycles called bio-geochemical cycles (literally, life-earth-chemical cycles), or **nutrient cycles**.

# Nutrient cycles

- Water cycle
- Carbon cycle
- Nitrogen cycle
- Phosphorus cycle
- Sulphur cycles

# Nutrient cycles

- As nutrients move through their bio-geochemical cycles, they may accumulate in certain portions of the cycles and remain there for different periods of time.
- These temporary storage sites such as the atmosphere, the oceans and other bodies of water, and underground deposits are called **reservoirs**.

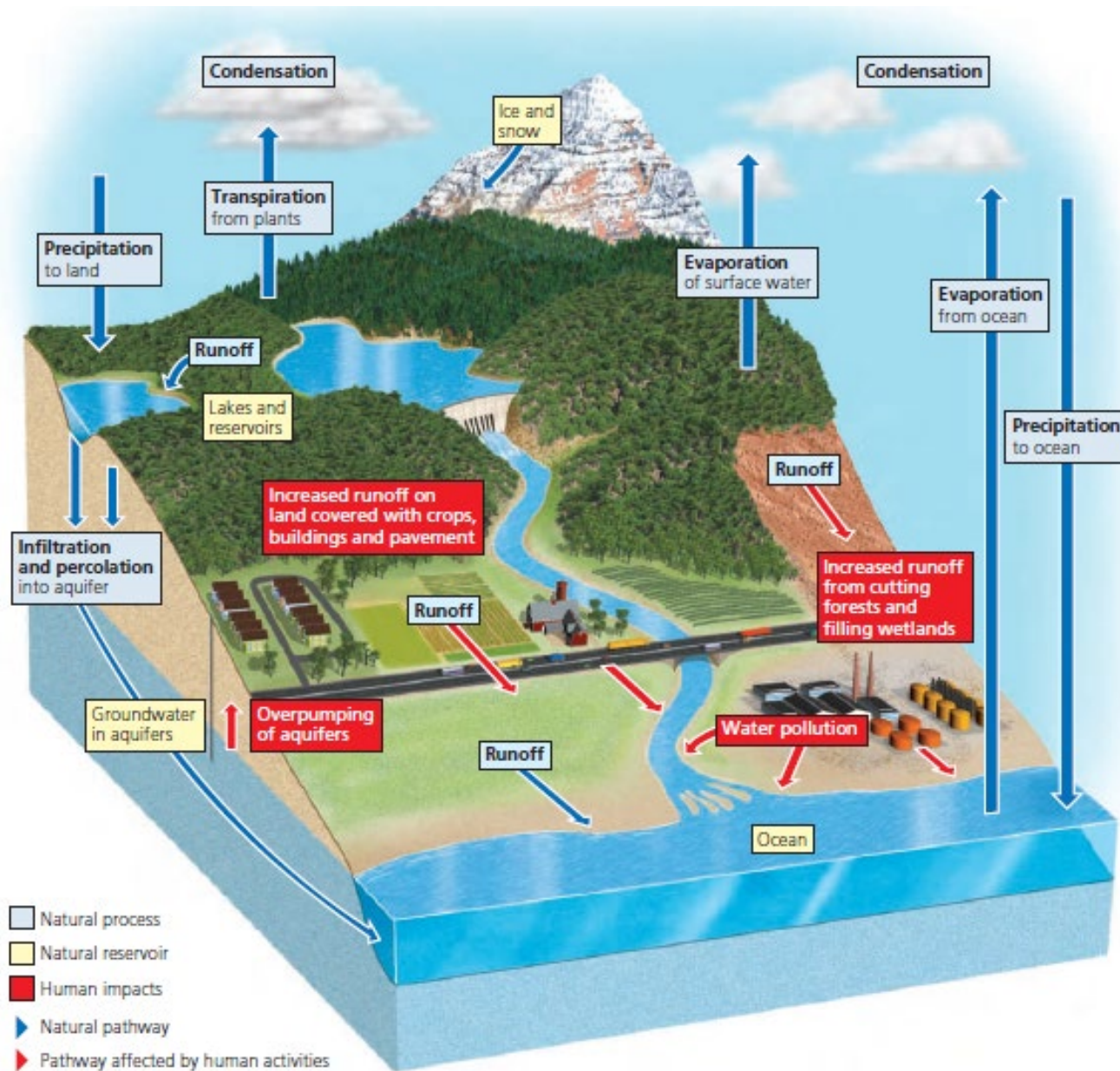
# Water cycle

**Water** is an amazing substance that is necessary for life on the earth.

The hydrologic cycle, or **water cycle**, collects, purifies, and distributes the earth's fixed supply of water,

## Why Water cycle important

Because water dissolves many nutrient compounds, it is a major medium for transporting nutrients within and between ecosystems.



# Water cycle

The water cycle is powered by energy from the sun and involves three major processes—

- Evaporation
- precipitation
- transpiration

# Water cycle

The water cycle is powered by energy from the sun and involves three major processes—

- Evaporation: Incoming solar energy causes evaporation of water from the earth's oceans, lakes, rivers, and soil. Evaporation changes liquid water into water vapor in the atmosphere
- Precipitation: gravity draws the water back to the earth's surface as precipitation (rain, snow, dew)
- Transpiration: water that reaches the atmosphere evaporates from the surfaces of plants, through a process called transpiration, and from the soil.

# Water cycle- various paths

Water returning to the earth's surface as precipitation takes various paths.

- Most precipitation falling on terrestrial ecosystems becomes *surface runoff*.
- This water flows into streams, which eventually carry water back to lakes and oceans, from which it can evaporate to repeat the cycle.
- Some precipitation is converted to ice that is stored in *glaciers*

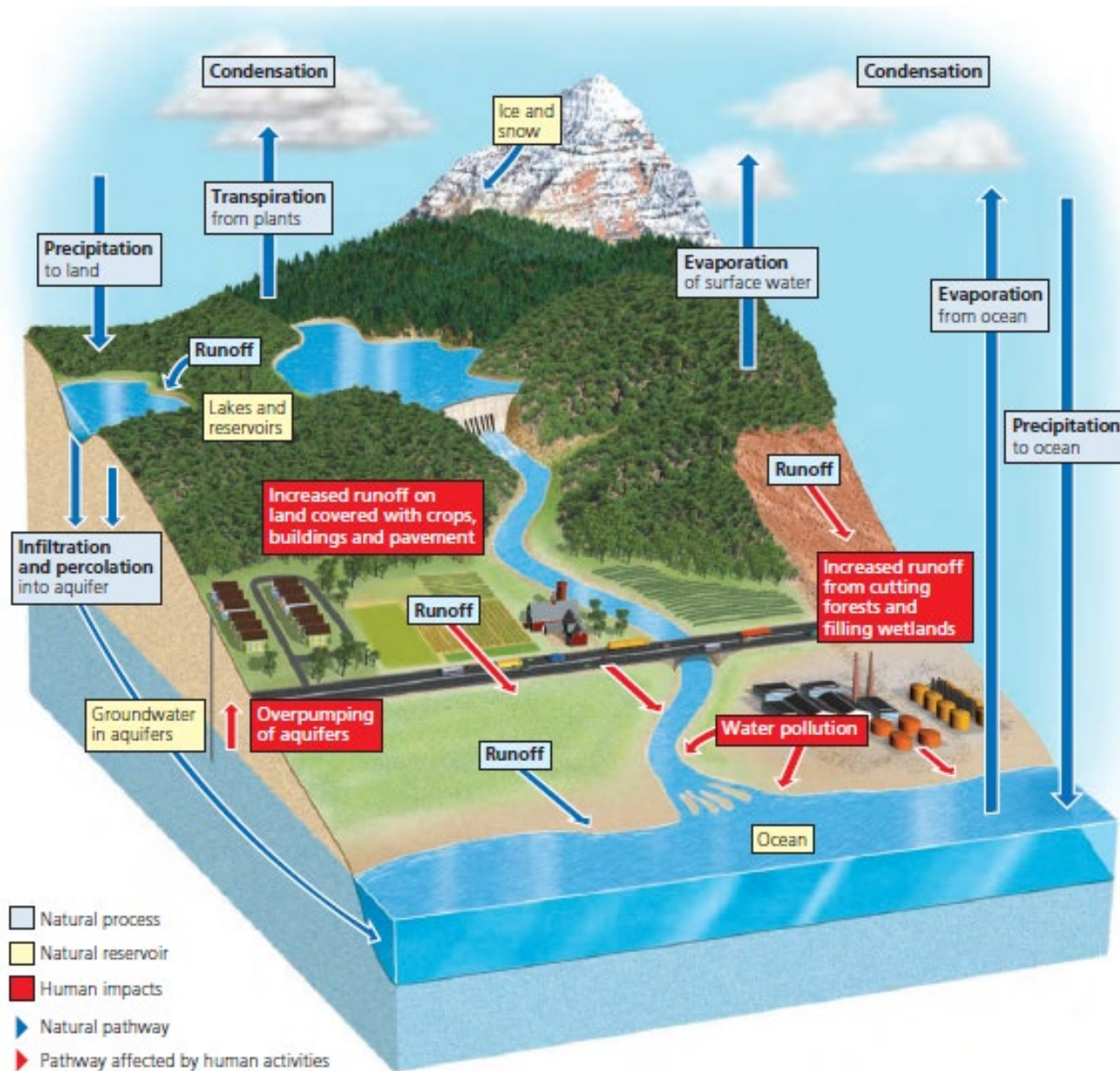
## Water cycle- various paths

- Some precipitation sinks through soil and permeable rock formations to underground layers of rock, sand, and gravel called *aquifers*, where it is stored as *groundwater*.
- Some surface water also seeps into the upper layers of soils where it is used by plants, and some evaporates from the soils back into the atmosphere.
- some of the water combines with carbon dioxide during photosynthesis to produce high-energy organic compounds such as carbohydrates.

## Water cycle- water availability

Only about 0.024% of the earth's vast water supply is available to humans and other species as liquid freshwater in accessible groundwater deposits and in lakes, rivers, and streams.

The rest is too salty for us to use, is stored as ice, or is too deep underground to extract at affordable prices using current technology.



# Affect of human activities on water cycle

We alter the water cycle in **three** major ways

- Consume excess of water
- Covering land with concrete
- Urban development with poor planning

# Affect of human activities on water cycle

We alter the water cycle in **three** major ways

1. We withdraw large quantities of freshwater from streams, lakes, and aquifers sometimes at rates faster than nature can replace it.
2. Mining, road building, and other activities cover much of the land with buildings, concrete, and asphalt.

This increases runoff, **reduces infiltration** that would normally recharge groundwater supplies, accelerates **topsoil erosion**, and increases the **risk of flooding**.

# Affect of human activities on water cycle

We alter the water cycle in **three** major ways

3. we also increase flooding when we fill wetlands for urban development (poor planning)

Left undisturbed, wetlands provide the natural service of flood control, acting like sponges to absorb and hold overflows of water from drenching rains or rapidly melting snow.

# Water cycle- Summary

Three processes in water cycle are

- Evaporation
- precipitation
- Transpiration

Human activities

- Withdrawing freshwater at faster rates
- Covering lands with concrete and buildings
- Urban development without proper planning

# Carbon cycle

# Carbon cycle

- **Carbon**-building block of carbohydrates, fats, DNA
- **Circulates**- biosphere, the atmosphere, and parts of the hydrosphere
- **Based on**- carbon dioxide ( $\text{CO}_2$ )

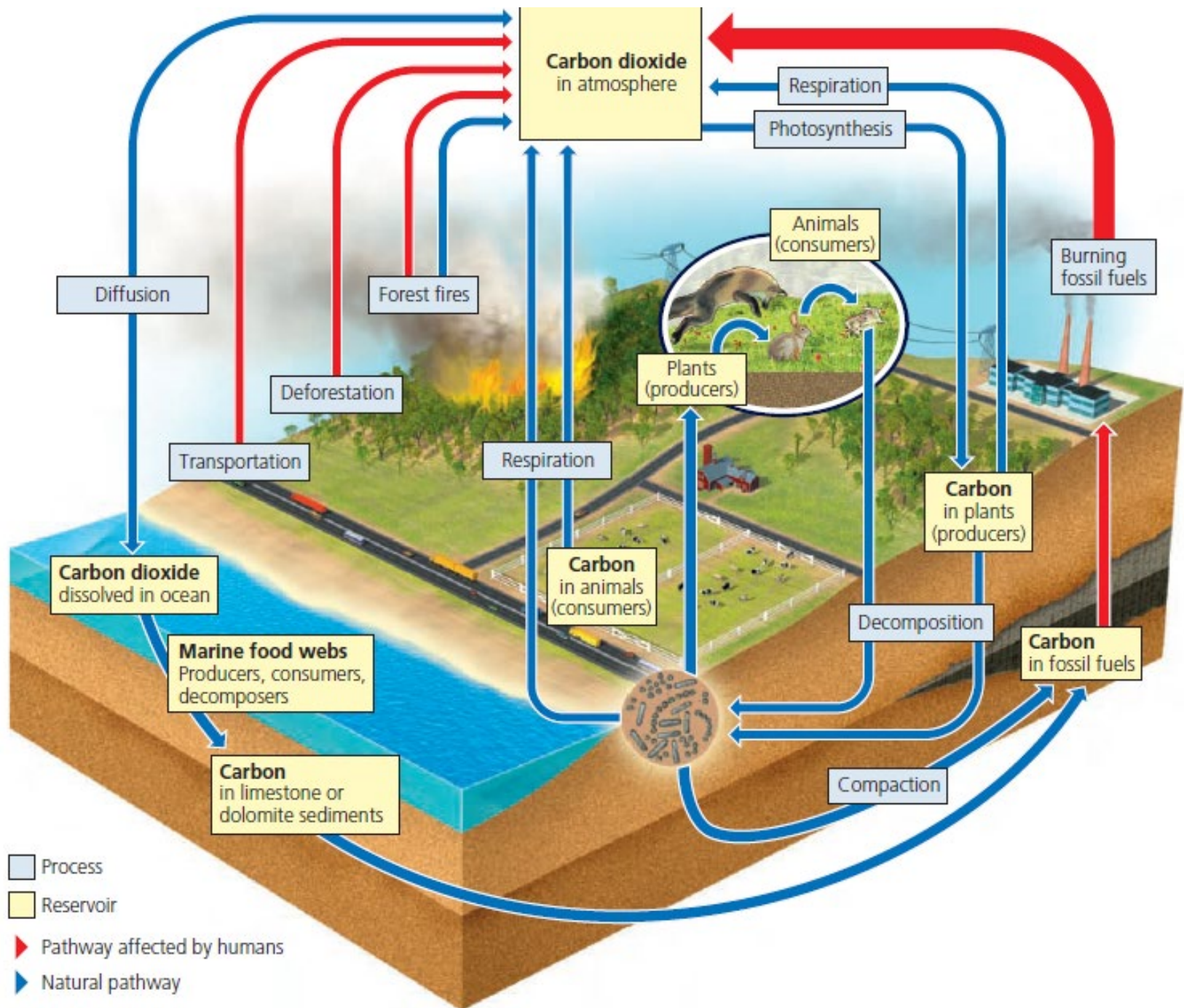
# Carbon cycle

- Carbon is the basic building block of the carbohydrates, fats, proteins, DNA, and other organic compounds necessary for life.
- Various compounds of carbon circulate through the biosphere, the atmosphere, and parts of the hydrosphere, in the carbon cycle
- The carbon cycle is based on carbon dioxide ( $\text{CO}_2$ ) gas, which makes up 0.039% of the volume of the earth's atmosphere

# Carbon cycle

- The carbon cycle is based on carbon dioxide ( $\text{CO}_2$ ) gas, which makes up 0.039% of the volume of the earth's atmosphere
- If the carbon cycle removes too much  $\text{CO}_2$  from the atmosphere, the atmosphere will cool,
- if it generates too much  $\text{CO}_2$ , the atmosphere will get warmer.

Thus, even slight changes in this cycle caused by natural or human factors can affect the earth's climate



# Carbon cycle

Main processes in carbon cycle

- Photosynthesis
- Aerobic respiration

This linkage between *photosynthesis* in producers and *aerobic respiration* in producers, consumers, and decomposers **circulates carbon** in the biosphere.

# Carbon cycle

## Decomposers role

- **On Land**-Decomposers release the carbon stored in the bodies of dead organisms on land back into the air as  $CO_2$
- **In ocean**-in water, decomposers release carbon that can be stored as insoluble carbonates in bottom sediment

Over millions of years, buried deposits of dead plant matter and bacteria are compressed between layers of sediment, where high pressure and heat convert them to carbon-containing fossil fuels such as coal, oil, and natural gas

# Carbon cycle

## Fossil fuels

- Over millions of years, buried deposits of dead plant matter and bacteria are compressed between layers of sediment, where high pressure and heat convert them to carbon-containing fossil fuels such as coal, oil, and natural gas
- This carbon is not released to the atmosphere as  $\text{CO}_2$  for recycling until these fuels are extracted and burned,
- In only a few hundred years, we have extracted and burned huge quantities of fossil fuels that took millions of years to form

# Carbon cycle

Human activities

## Fossil fuels burning

We are altering the carbon cycle mostly by adding large amounts of carbon dioxide to the atmosphere, when we burn carbon-containing fossil fuels (example: coal to produce electricity).

# Carbon cycle

## Human activities

### Clearing vegetation

We also alter the cycle by clearing carbon-absorbing vegetation from forests (especially tropical forests) faster than it can grow back

# Carbon cycle

## Human activities

### Carbon footprint

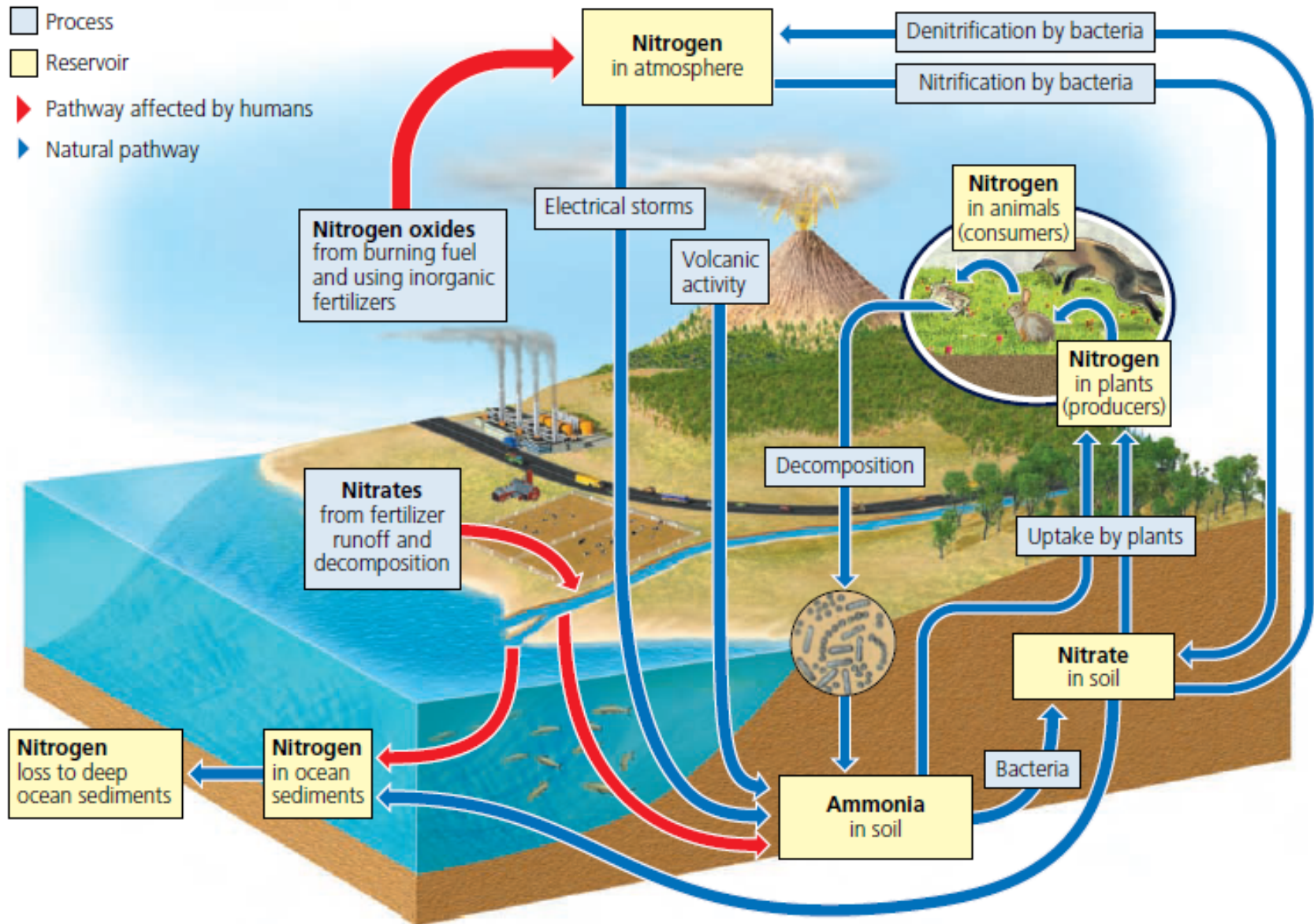
- Human activities are altering both the rate of energy flow and the cycling of nutrients within the carbon cycle
- In other words, humanity has a large and growing **carbon footprint** that makes up a significant part of our overall ecological footprint

# The Nitrogen Cycle

# Nitrogen Cycle

- Nitrogen is a crucial component of proteins, many vitamins, and nucleic acids such as DNA
- The major reservoir for nitrogen is the atmosphere. Chemically unreactive nitrogen gas ( $N_2$ ) makes up 78% of the volume of the atmosphere.
- However,  $N_2$  cannot be absorbed and used directly as a nutrient by multicellular plants or animals

# Nitrogen Cycle



# Nitrogen Cycle steps

Nitrogen fixation

Nitrification

Ammonification

Denitrification

# Nitrogen Cycle steps

## Nitrogen fixation

Nitrogen fixation can happen in two processes

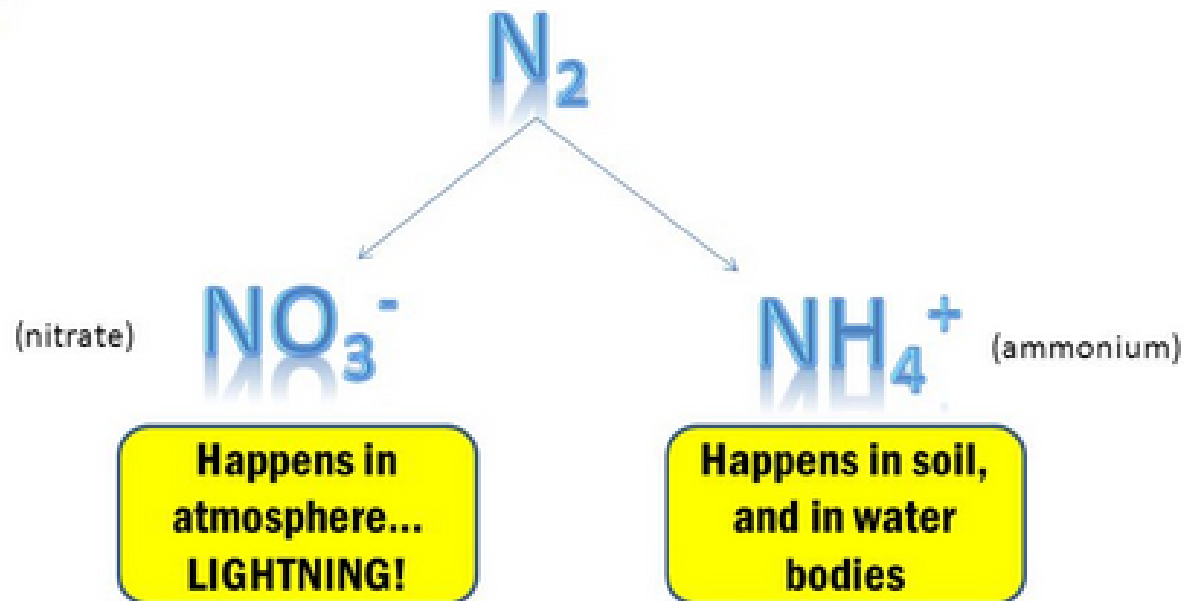
1. Lightning
2. Nitrogen fixing bacteria

# Nitrogen Cycle steps

## #1 – Nitrogen Fixation



- This is a process that changes  $N_2$  into forms that plants can use!



# Nitrogen Cycle steps

## Nitrogen fixation

### Lightning

- lightning breaks nitrogen molecules and enables their atoms to combine with oxygen in the air forming nitrogen oxides.
- Nitrogen oxides dissolve in rain, forming nitrates, that are carried to the earth. Atmospheric nitrogen fixation probably contributes some 5- 8% of the total nitrogen fixed.

Nitrogen  Nitrates

# Nitrogen Cycle steps

## Nitrogen fixation

### Nitrogen fixing bacteria

- Takes place in aquatic systems, in soil, and in the roots of some plants.
- In nitrogen fixation, specialized bacteria in soil as well as bluegreen algae (cyanobacteria) in aquatic environments combine gaseous  $N_2$  with hydrogen to make ammonia ( $NH_3$ ).
- Some of the ammonia is converted to ammonium ions ( $NH_4^+$ ) that plants can use as a nutrient.

Nitrogen  Ammonia and ammonium ions

# Nitrogen Cycle steps

## nitrification

- In this process, specialized soil bacteria convert most of the  $\text{NH}_3$  and  $\text{NH}_4^+$  in soil to nitrate ions ( $\text{NO}_3^-$ ), which are easily taken up by the roots of plants.
- Nitrogen get transferred from producers to consumers and decomposers

Ammonia and  
ammonium ions



Nitrates

# Nitrogen Cycle steps

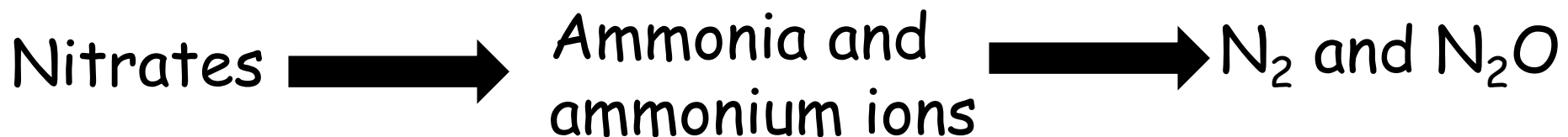
## Ammonification

- Plants and animals return nitrogen-rich organic compounds to the environment as both wastes and cast-off particles of tissues
- In ammonification, vast armies of specialized decomposer bacteria convert this detritus into simpler nitrogen containing inorganic compounds such as ammonia ( $\text{NH}_3$ ) and water-soluble salts containing ammonium ions ( $\text{NH}_4^+$ ).

# Nitrogen Cycle steps

## denitrification

- Specialized bacteria in waterlogged soil and in the bottom sediments of lakes, oceans convert  $\text{NH}_3$  and  $\text{NH}_4^+$  back into nitrate ions, and then into nitrogen gas ( $\text{N}_2$ ) and nitrous oxide gas ( $\text{N}_2\text{O}$ ).
- These gases are released to the atmosphere to begin the nitrogen cycle again.



# Human activities affect on nitrogen cycle

## 1. Nitric oxide release

- we add large amounts of nitric oxide (NO) into the atmosphere when  $\text{N}_2$  and  $\text{O}_2$  combine as we burn any fuel at high temperatures in vehicles.
- this gas can be converted to nitrogen dioxide gas ( $\text{NO}_2$ ) and nitric acid vapor ( $\text{HNO}_3$ ), which can return to the earth's surface as damaging acid deposition, commonly called **acid rain**.

# Human activities affect on nitrogen cycle

## 2. Nitrous oxide release

- we add nitrous oxide ( $\text{N}_2\text{O}$ ) to the atmosphere through the action of anaerobic bacteria on commercial inorganic fertilizer or organic animal manure applied to the soil.
- This **greenhouse gas** can warm the atmosphere and deplete stratospheric ozone,

# Human activities affect on nitrogen cycle

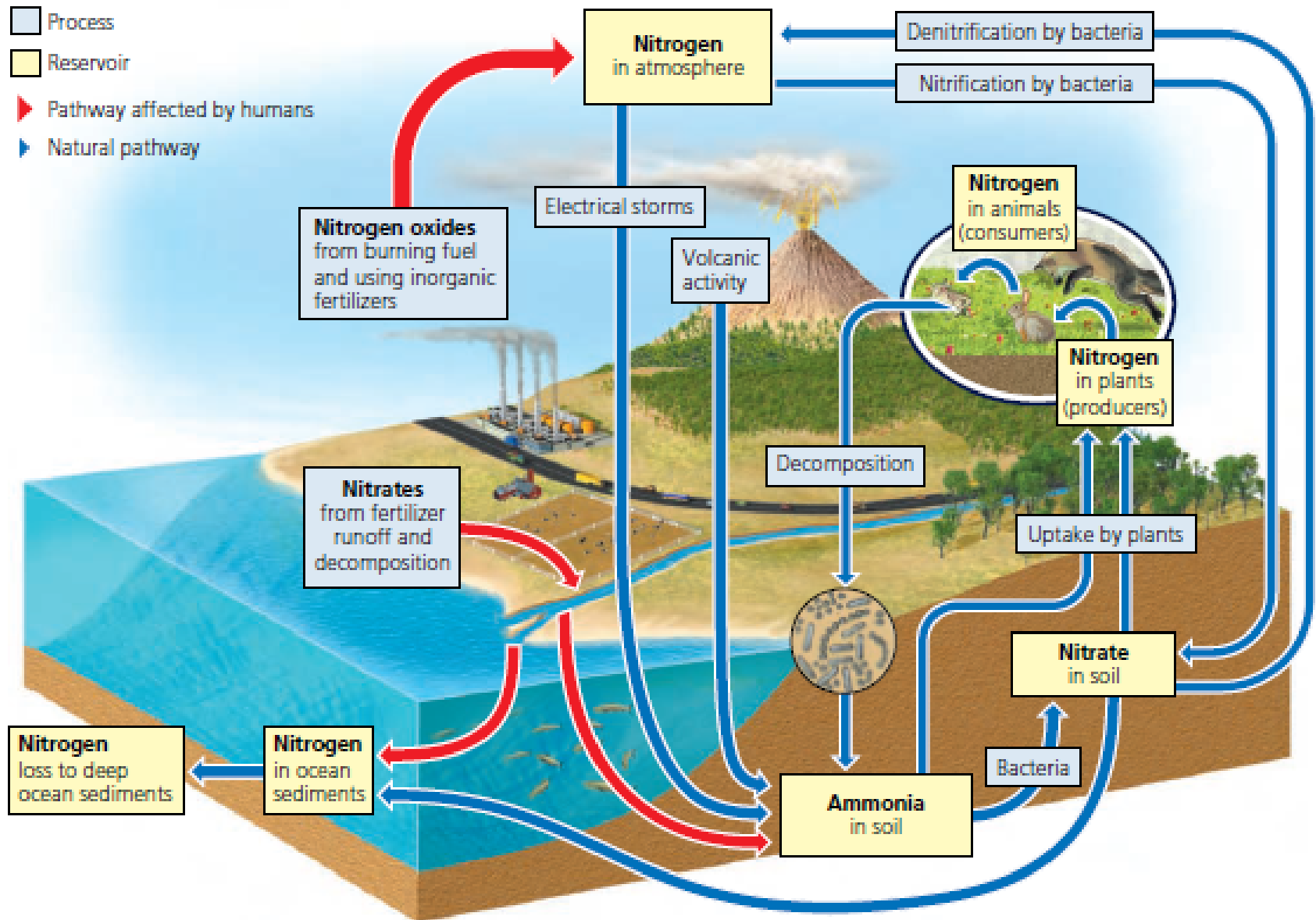
## 3. excess nitrates ( $\text{NO}_3^-$ ) in water

- We upset the nitrogen cycle in aquatic ecosystems by adding excess nitrates ( $\text{NO}_3^-$ ) to bodies of water through agricultural runoff of fertilizers and through discharges from municipal sewage systems.
- This can cause excess growth of algae

# Human activities affect on nitrogen cycle

## Removal of nitrogen from top soil

- we remove nitrogen from topsoil when we harvest nitrogen-rich crops, irrigate crops (washing nitrates out of the soil), and burn or clear grasslands and forests before planting crops.



# Nitrogen cycle- Summary

Three processes in nitrogen cycle are

Nitrogen fixation

Nitrification

Ammonification

Denitrification

Human activities

- Nitric oxide release
- Nitrous oxide release
- excess nitrates ( $\text{NO}_3^-$ ) in water

# Phosphorus Cycle

# Phosphorus Cycle

- Compounds of phosphorous (P) circulate through water, the earth's crust, and living organisms in the **phosphorus cycle**
- Most of these compounds contain phosphate ions ( $\text{PO}_4^{3-}$ ), which are a combination of phosphorus and oxygen and which serve as an important nutrient.
- In contrast to the cycles of water, carbon, and nitrogen, the phosphorus cycle **does not include the atmosphere.**

# Phosphorus Cycle

phosphates are a component of biologically important molecules

- Nucleic acids
- ADP and ATP
- bones and teeth

# Phosphorus Cycle

- The phosphorus cycle is slow compared to the water, carbon, and nitrogen cycles.

rocks containing **Phosphate**



Water

Plants



Consumers



Decomposers

# Phosphorus Cycle

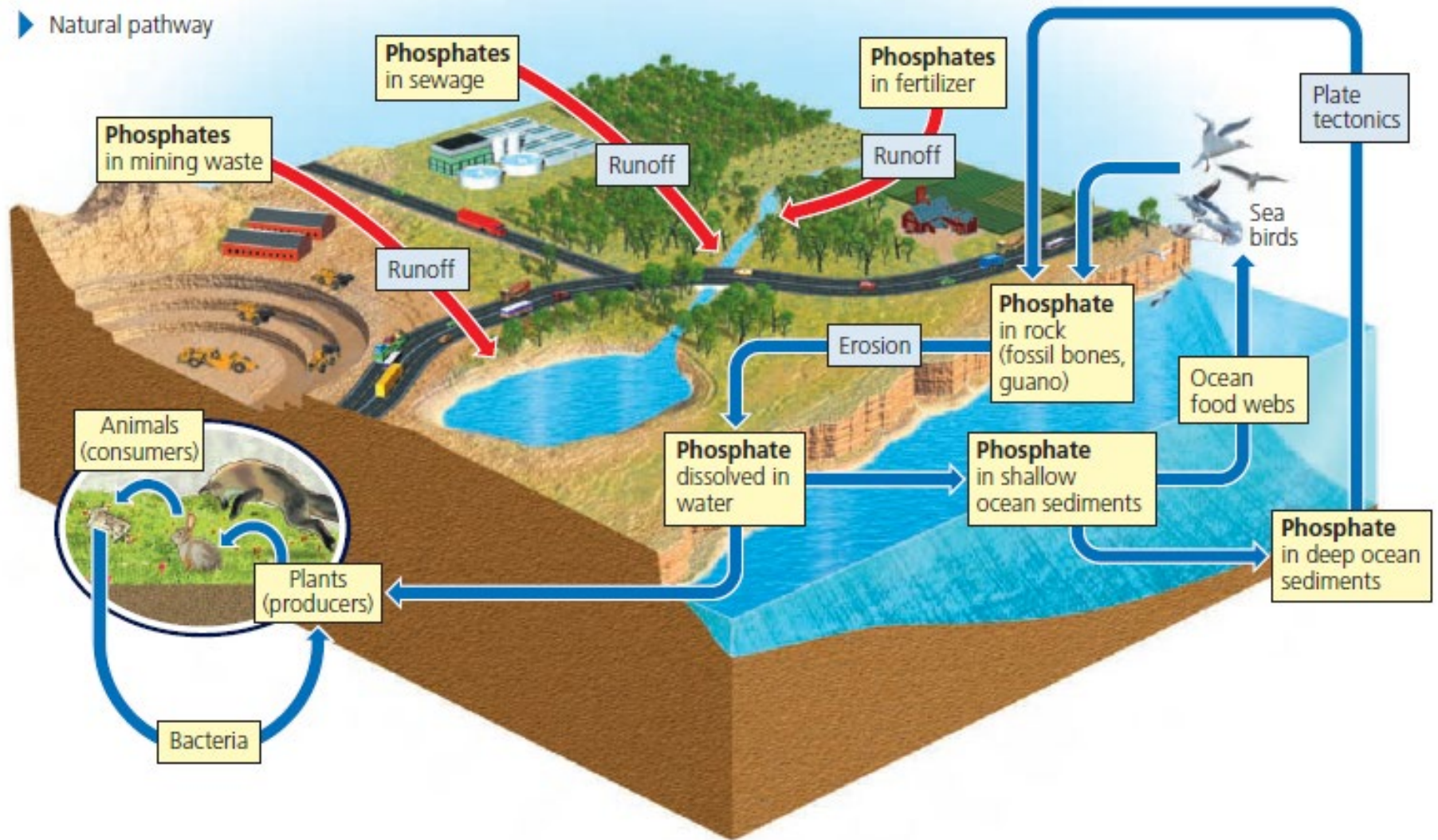
- The **lack** of phosphate often **limits plant growth** on land
- Most soils contain little phosphate
- phosphorus (as phosphate salts mined from the earth) is applied to the soil as a fertilizer.

Process

Reservoir

▶ Pathway affected by humans

▶ Natural pathway



# Affect of human activities on Phosphorus Cycle

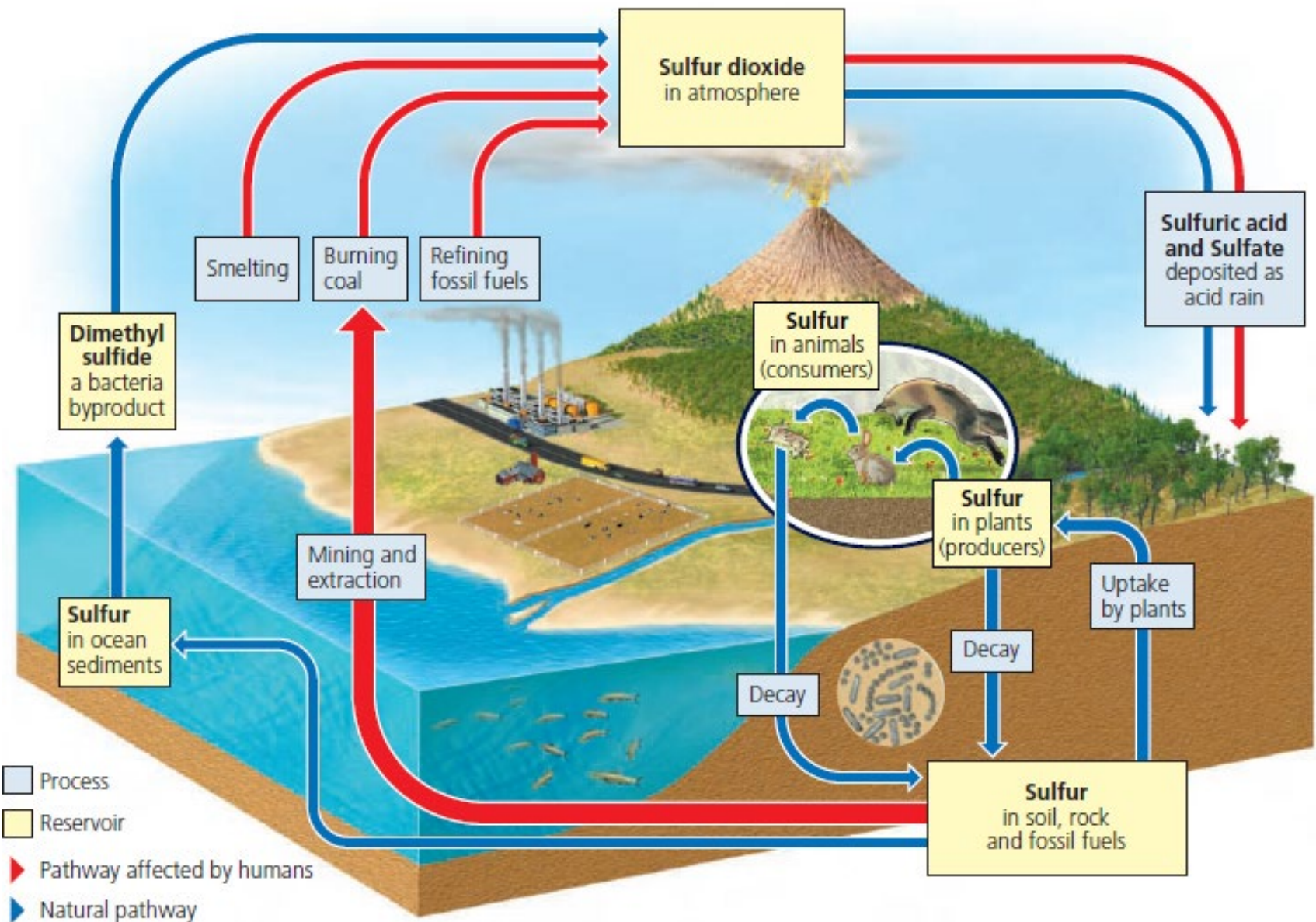
## Human activities

- removing large amounts of phosphate from the earth to make fertilizer.
- Reducing phosphate levels in tropical soils by clearing forests

## Affects

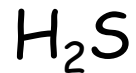
- Phosphates in large quantities enter into streams, lakes, and oceans; there they stimulate the growth of producers such as algae and various aquatic plants.

# The Sulfur Cycle



# The Sulfur Cycle

- Much of the earth's sulfur is stored underground in rocks and minerals and in the form of sulfate ( $\text{SO}_4^{2-}$ ) salts buried deep under ocean sediments.
- Sulfur also enters the atmosphere in the form of



- Plant roots absorb sulfate ions and incorporate the sulfur as an essential component of many proteins.

# The Sulfur Cycle

1. Hydrogen sulfide ( $\text{H}_2\text{S}$ )—a colorless, highly poisonous gas with a rotten-egg smell—is released from active volcanoes

from organic matter broken down by anaerobic decomposers in flooded swamps

2. Sulfur dioxide ( $\text{SO}_2$ ), a colorless and suffocating gas, also comes from volcanoes

3. Particles of sulfate ( $\text{SO}_4^{2-}$ ) salts, such as ammonium sulfate, enter the atmosphere from sea spray, dust storms, and forest fires.

# The Sulfur Cycle

- In the oxygen-deficient environments of flooded soils, freshwater wetlands, specialized bacteria convert sulfate ions to sulfide ions ( $S_2^-$ ).
- The sulfide ions can then react with metal ions to form insoluble metallic sulfides, which are deposited as rock or metal ores (often extracted by mining and converted to various metals), and the cycle continues.

# The Sulfur Cycle

## Dimethyl sulfide role

- Certain marine algae produce large amounts of **volatile** dimethyl sulfide, or DMS ( $\text{CH}_3\text{SCH}_3$ ).
- Tiny droplets of **DMS serve as nuclei** for the condensation of water into droplets found in clouds.
- In this way, changes in DMS emissions can affect cloud cover and climate.

# The Sulfur Cycle

## Dimethyl sulfide- Acid rain

In the atmosphere, DMS is converted to sulfur dioxide, some of which in turn is converted to sulfur trioxide gas ( $\text{SO}_3$ ) and to tiny droplets of sulfuric acid ( $\text{H}_2\text{SO}_4$ )



These droplets fall to the earth as components of *acid deposition*, which along with other air pollutants can harm trees and aquatic life.

# The Sulfur Cycle

## Dimethyl sulfide- Acid rain

- DMS also reacts with other atmospheric chemicals such as ammonia to produce tiny particles of sulfate salts.



These particles fall to the earth as components of *acid deposition*, which along with other air pollutants can harm trees and aquatic life.

## The Sulfur Cycle-Human activities

- Human activities have affected the sulfur cycle primarily by releasing large amounts of sulfur dioxide ( $\text{SO}_2$ ) into the atmosphere

We release sulfur to the atmosphere in three ways.

1. we burn sulfur-containing coal
2. we refine sulfur-containing oil (petroleum) to make gasoline
3. Third, we extract metals such as copper, lead, and zinc from sulfur-containing compounds in rocks that are mined for these metals.

## The Sulfur Cycle-Human activities

Once the sulfur is in the atmosphere,  $\text{SO}_2$  is converted to droplets of sulfuric acid ( $\text{H}_2\text{SO}_4$ ) and particles of sulfate ( $\text{SO}_4^{2-}$ ) salts, which return to the earth as acid deposition.